

Cloud Computing

Lab-2

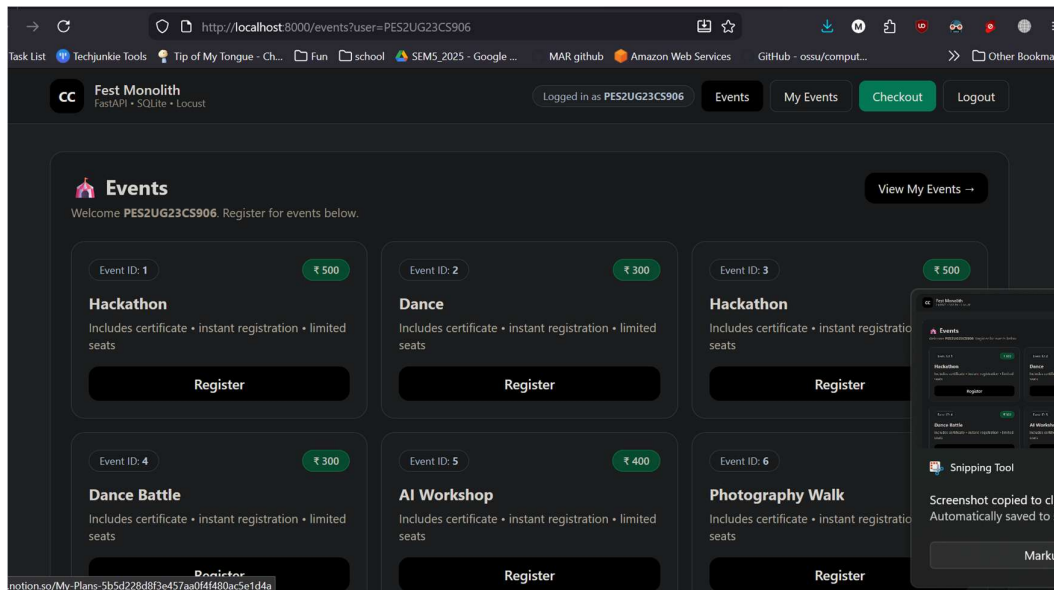
Name: Monisha Sharma

SRN: PES2UG23CS906

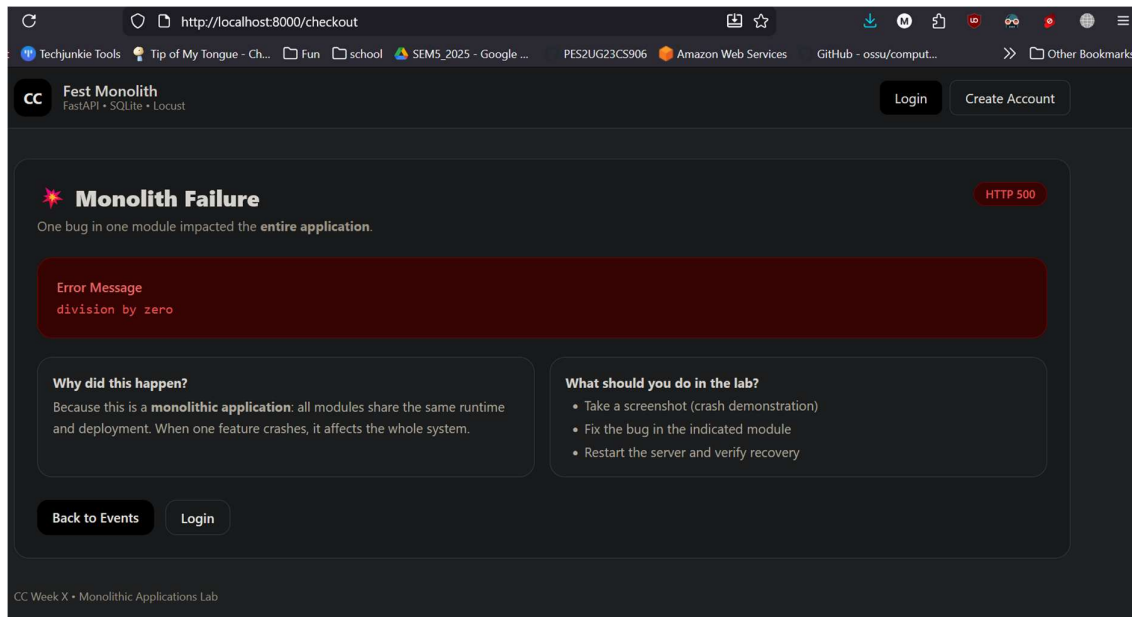
Class: 6F

Date: 20/01/2026

Events page



Checkout crash



→ ↺

🔍 http://localhost:8000/checkout

Task List Techjunkie Tools Tip of My Tongue - Ch... Fun school SEM5_2025 - Google ...

CC Fest Monolith

FastAPI • SQLite • Locust

🔴 Monolith Failure

One bug in one module impacted the **entire application**.

Error Message

division by zero

Why did this happen?

Because this is a **monolithic application**: all modules share the same runtime and deployment. When one feature crashes, it affects the whole system.

Back to Events

Login

CC Week X • Monolithic Applications Lab

```
File "/home/mona/PES2UG23CS906/CC Lab-2/.venv/lib/python3.10/site-packages/starlette/_exception_handler.py", line 42, in wrapped_app
    await app(scope, receive, sender)
File "/home/mona/PES2UG23CS906/CC Lab-2/.venv/lib/python3.10/site-packages/fastapi/routing.py", line 101, in app
    response = await f(request)
File "/home/mona/PES2UG23CS906/CC Lab-2/.venv/lib/python3.10/site-packages/fastapi/routing.py", line 355, in app
    raw_response = await run_endpoint_function(
File "/home/mona/PES2UG23CS906/CC Lab-2/.venv/lib/python3.10/site-packages/fastapi/routing.py", line 245, in run_endpoint_function
    return await run_in_threadpool(dependant.call, **values)
File "/home/mona/PES2UG23CS906/CC Lab-2/.venv/lib/python3.10/site-packages/starlette/concurrency.py", line 32, in run_in_threadpool
    return await anyio.to_thread.run_sync(func)
File "/home/mona/PES2UG23CS906/CC Lab-2/.venv/lib/python3.10/site-packages/anyio/to_thread.py", line 63, in run_sync
    return await get_async_backend().run_sync_in_worker_thread(
File "/home/mona/PES2UG23CS906/CC Lab-2/.venv/lib/python3.10/site-packages/anyio/_backends/_asyncio.py", line 2502, in run_sync_in_worker_thread
    return await future
File "/home/mona/PES2UG23CS906/CC Lab-2/.venv/lib/python3.10/site-packages/anyio/_backends/_asyncio.py", line 986, in run
    result = context.run(func, *args)
File "/home/mona/PES2UG23CS906/CC Lab-2/main.py", line 113, in checkout
    total = checkout_logic()
File "/home/mona/PES2UG23CS906/CC Lab-2/checkout/_init_.py", line 10, in checkout_logic
    1 / 0
ZeroDivisionError: division by zero
```

Fixed

You are signed in as PES2UG23CS906

Profile | MyCourses

Monolithic Applications - Monolith

CC Fest Monolith

CC Fest Monolith

PES2UG23CS906 - Google

+

▼

← → ↺

🔍 http://localhost:8000/checkout

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CC Fest Monolith

FastAPI • SQLite • Locust

Login Create Account

🏠 Checkout

This route is used to demonstrate a monolith crash + optimization.

Total Payable

₹ 6600

✓ After fixing + optimizing checkout logic, re-run Locust and compare results.

What you should observe

- One buggy feature can crash the entire monolith.
- Inefficient loops cause high response times under load.
- Optimization improves performance but architecture still scales as one unit.

Next Lab: Split this monolith into Microservices (Events / Registration / Checkout).

CC Week X • Monolithic Applications Lab

→ ↺

🔒

📄

http://localhost:8000/checkout

Task ListTechjunkie Tools🔔 Tip of My Tongue - Ch...📁 Fun📁 school🔥 SEM5_2025 - Google ...

CC

Fest Monolith

FastAPI • SQLite • Locust

🛒 Checkout

This route is used to demonstrate a monolith crash + optimization.

Total Payable

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✅ After fixing + optimizing checkout logic, re-run Locust and compare results.

CC Week X • Monolithic Applications Lab

```
File "/home/mona/PES2UG23CS906/CC Lab-2/.venv/lib/python3.10/s
ite-packages/fastapi/routing.py", line 245, in run_endpoint_func
tion
    return await run_in_threadpool(dependant.call, **values)
File "/home/mona/PES2UG23CS906/CC Lab-2/.venv/lib/python3.10/s
ite-packages/starlette/concurrency.py", line 32, in run_in_threa
dpool
    return await anyio.to_thread.run_sync(func)
File "/home/mona/PES2UG23CS906/CC Lab-2/.venv/lib/python3.10/s
ite-packages/anyio/to_thread.py", line 63, in run_sync
    return await get_async_backend().run_sync_in_worker_thread(
File "/home/mona/PES2UG23CS906/CC Lab-2/.venv/lib/python3.10/s
ite-packages/anyio/_backends/_asyncio.py", line 2502, in run_syn
c_in_worker_thread
    return await future
File "/home/mona/PES2UG23CS906/CC Lab-2/.venv/lib/python3.10/s
ite-packages/anyio/_backends/_asyncio.py", line 986, in run
    result = context.run(func, *args)
File "/home/mona/PES2UG23
checkout
    total = checkout_logic(
File "/home/mona/PES2UG23
line 10, in checkout_logic
    1 / 0
ZeroDivisionError: division
WARNING: StatReload detect
Reloading...
INFO: Shutting down
INFO: Waiting for appli
INFO: Application shutd
INFO: Finished server p
INFO: Started server pr
INFO: Waiting for appli
INFO: Application start
INFO: 127.0.0.1:46630 -
```

Checkout

Total Payable

₹ 6600

What you should observe

- ✓ Total Payable should be ₹ 6600
- ✓ After fixing + optimizing checkout logic, re-run Locust and compare results.

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

LOCUST

← → ↺

🔒

📄

http://localhost:8089

Task ListTechjunkie Tools🔔 Tip of My Tongue - Ch...📁 Fun📁 school🔥 SEM5_2025 - Google ...mar githubAmazon Web ServicesGitHub - ossu/comput...Other Bookmarks

🦋 LOCUST

Host

http://localhost:8000

Status

STOPPED

RPS

0.7

Failures

0%

NEW

RESET

⚙️

STATISTICS

CHARTS

FAILURES

EXCEPTIONS

CURRENT RATIO

DOWNLOAD DATA

LOGS

⏸

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/checkout	21	0	7	8	33	7.99	4	33	2797	0.7	0
Aggregated		21	0	7	8	33	7.99	4	33	2797	0.7	0

```
mona@MonasLaptop X mona@MonasLaptop X + v - □ X
[2026-01-20 14:50:20,884] MonasLaptop/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 14:50:20,885] MonasLaptop/INFO/locust.main: Starting web interface at http://0.0.0.0:8089, press enter to open your default browser.
[2026-01-20 14:51:53,064] MonasLaptop/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 14:51:53,070] MonasLaptop/INFO/locust.runners: All users spawned: {"CheckoutUser": 1} (1 total users)
KeyboardInterrupt
2026-01-20T09:22:40Z
[2026-01-20 14:52:40,530] MonasLaptop/INFO/locust.main: Shutting down (exit code 0)
Type      Name  # reqs      # fails |      Avg      Min      Max      Med
ed |   req/s  failures/s
-----|-----|-----|-----|-----|-----|-----|
GET      /checkout    21      0(0.00%) |      7      4     32
7 |      0.72      0.00
-----|-----|-----|-----|-----|-----|
-----|-----|-----|-----|-----|-----|
2      Aggregated    21      0(0.00%) |      7      4      3
7 |      0.72      0.00

Response time percentiles (approximated)
Type      Name      50%      66%      75%      80%      90%      95%      98%
99%      99.9% 99.99% 100% # reqs
-----|-----|-----|-----|-----|-----|-----|-----|
GET      /checkout      7      7      8      8      8      8
33      33      33      33      33      21
-----|-----|-----|-----|-----|-----|-----|
-----|-----|-----|-----|-----|-----|
      Aggregated      7      7      8      8      8      8
33      33      33      33      33      21

(.venv) mona@MonasLaptop:~/PES2UG23CS906/CC Lab-2$
```

LOCUST

STATISTICSCHARTSFAILURES EXCEPTIONSCURRENT

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Avg (ms)
GET	/checkout	21	0	7	8	33	7.9
Aggregated		21	0	7	8	33	7.9

ABOUT

mona@MonasLaptop

```
[2026-01-20 14:50:20,884] MonasLaptop/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 14:50:20,885] MonasLaptop/INFO/locust.main: Starting web interface at http://0.0.0.0:8089, press enter to open your default browser.
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[2026-01-20 14:51:53,070] MonasLaptop/INFO/locust.runners: All users spawned: {"CheckoutUser": 1} (1 total users)
KeyboardInterrupt
2026-01-20T09:22:40Z
[2026-01-20 14:52:40,530] MonasLaptop/INFO/locust.main: Shutting down (exit code 0)
Type Name # reqs # fails | Avg Min Max Med | req/s failures/s
-----|-----|-----|-----|-----|-----|
GET /checkout 21 0(0.00%) | 7 4 32 7 | 0.72 0.00
-----|-----|-----|-----|-----|
2 Aggregated 21 0(0.00%) | 7 4 3 7 | 0.72 0.00

Response time percentiles (approximated)
Type Name 50% 66% 75% 80% 90% 95% 98%
-----|-----|-----|-----|-----|-----|
GET /checkout 7 7 8 8 8 8 8
33 33 33 33 21
-----|-----|-----|-----|-----|
Aggregated 7 7 8 8 8 8 8
33 33 33 33 21

(.venv) mona@MonasLaptop:~/PES2UG23CS906/CC Lab-2$
```

RESTART

LOCUST

STATISTICSCHARTSFAILURES EXCEPTIONSCURRENT RATIO DOWNLOAD DATA LOGS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/checkout	21	0	6	7	14	6.18	4	14	2797	0.8	0
Aggregated		21	0	6	7	14	6.18	4	14	2797	0.8	0

NEW RESET

mona@MonasLaptop

```
[2026-01-20 14:47:28,913] MonasLaptop/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 14:47:28,913] MonasLaptop/INFO/locust.main: Starting web interface at http://0.0.0.0:8089, press enter to open your default browser.
[2026-01-20 14:48:04,196] MonasLaptop/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 14:48:04,197] MonasLaptop/INFO/locust.runners: All users spawned: {"CheckoutUser": 1} (1 total users)
KeyboardInterrupt
2026-01-20T09:19:03Z
[2026-01-20 14:49:03,336] MonasLaptop/INFO/locust.main: Shutting down (exit code 0)
Type Name # reqs # fails | Avg Min Max Med | req/s failures/s
-----|-----|-----|-----|-----|-----|
GET /checkout 21 0(0.00%) | 6 3 13 6 | 0.72 0.00
-----|-----|-----|-----|-----|
3 Aggregated 21 0(0.00%) | 6 3 1 6 | 0.72 0.00

Response time percentiles (approximated)
Type Name 50% 66% 75% 80% 90% 95% 98%
-----|-----|-----|-----|-----|-----|
GET /checkout 6 6 6 6 7 7 7
14 14 14 14 21
-----|-----|-----|-----|-----|
Aggregated 6 6 6 6 7 7 7
14 14 14 14 21

(.venv) mona@MonasLaptop:~/PES2UG23CS906/CC Lab-2$
```

LOCUST

STATISTICSCHARTSFAILURES EXCEPTIONSCURRENT

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Avg (ms)
GET	/checkout	21	0	6	7	14	6.1
Aggregated		21	0	6	7	14	6.1

ABOUT

mona@MonasLaptop

```
[2026-01-20 14:47:28,913] MonasLaptop/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 14:47:28,913] MonasLaptop/INFO/locust.main: Starting web interface at http://0.0.0.0:8089, press enter to open your default browser.
[2026-01-20 14:48:04,196] MonasLaptop/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 14:48:04,197] MonasLaptop/INFO/locust.runners: All users spawned: {"CheckoutUser": 1} (1 total users)
KeyboardInterrupt
2026-01-20T09:19:03Z
[2026-01-20 14:49:03,336] MonasLaptop/INFO/locust.main: Shutting down (exit code 0)
Type Name # reqs # fails | Avg Min Max Med | req/s failures/s
-----|-----|-----|-----|-----|-----|
GET /checkout 21 0(0.00%) | 6 3 13 6 | 0.72 0.00
-----|-----|-----|-----|-----|
3 Aggregated 21 0(0.00%) | 6 3 1 6 | 0.72 0.00

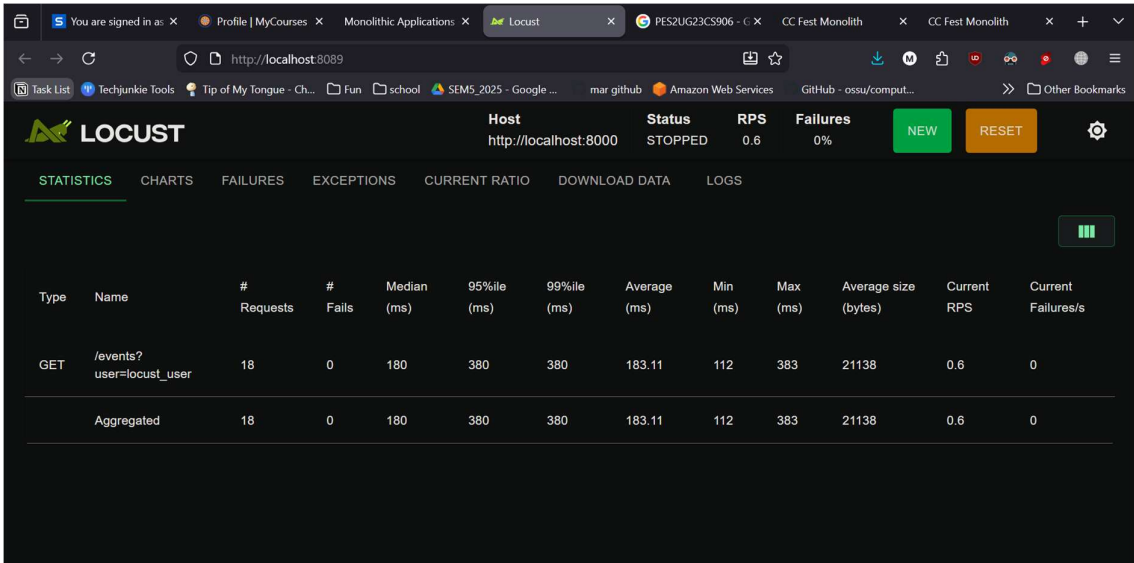
Response time percentiles (approximated)
Type Name 50% 66% 75% 80% 90% 95% 98%
-----|-----|-----|-----|-----|-----|
GET /checkout 6 6 6 6 7 7 7
14 14 14 14 21
-----|-----|-----|-----|-----|
Aggregated 6 6 6 6 7 7 7
14 14 14 14 21

(.venv) mona@MonasLaptop:~/PES2UG23CS906/CC Lab-2$
```

RPS WENT FROM 0.7 TO 0.8

ART WENT FROM 7.99 TO 6.18

Part 7




```
mona@MonasLaptop x mona@MonasLaptop x + v - □ X
[2026-01-20 14:57:10,441] MonasLaptop/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 14:57:10,442] MonasLaptop/INFO/locust.main: Starting web interface at http://0.0.0.0:8089, press enter to open your default browser.
[2026-01-20 14:58:02,227] MonasLaptop/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 14:58:02,236] MonasLaptop/INFO/locust.runners: All users spawned: {"EventsUser": 1} (1 total users)
KeyboardInterrupt
2026-01-20T09:30:07Z
[2026-01-20 15:00:07,832] MonasLaptop/INFO/locust.main: Shutting down (exit code 0)
Type      Name      # reqs      # fails |      Avg      Min      Max      M
ed |      req/s      failures/s
-----|-----|-----|-----|-----|-----|-----|
GET      /events?user=locust_user      18      0(0.00%) |      183
112      383      180 |      0.62      0.00
-----|-----|-----|-----|-----|-----|
-----|-----|-----|-----|-----|-----|
3      180 |      0.62      0.00      183      112      38

Response time percentiles (approximated)
Type      Name      50%      66%      75%      80%      90%      95%      98%
99%      99.9%      99.99%      100% # reqs
-----|-----|-----|-----|-----|-----|-----|-----|
GET      /events?user=locust_user      180      190      200      210
250      380      380      380      380      380      18
-----|-----|-----|-----|-----|-----|-----|
-----|-----|-----|-----|-----|-----|
380      Aggregated      180      190      200      210      250      380
380      380      380      380      380      18

(.venv) mona@MonasLaptop:~/PES2UG23CS906/CC Lab-2$
```

FileEditViewHistoryBookmarksProfilesToolsHelp

mona@MonasLapto

mona@MonasLapto

http://localhost:8089

Task ListTechjunkie ToolsTip of My Tongue - Ch...FunSchoolOther Bookmarks

LOCUST

STATISTICSCHARTSFAILURESEXCEPTIONSCURREN

STATISTICS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)
GET	/events?user=locust_user	18	0	180	380	380
Aggregated		18	0	180	380	380

ABOUT

[2026-01-20 14:57:10,441] MonasLaptop/INFO/locust.main: Starting Locust 2.43.1

[2026-01-20 14:57:10,442] MonasLaptop/INFO/locust.main: Starting web interface at http://0.0.0.0:8089, press enter to open your default browser.

[2026-01-20 14:58:02,227] MonasLaptop/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second

[2026-01-20 14:58:02,236] MonasLaptop/INFO/locust.runners: All users spawned: {"EventsUser": 1} (1 total users)

KeyboardInterrupt

2026-01-20T09:30:07Z

[2026-01-20 15:00:07,832] MonasLaptop/INFO/locust.main: Shutting down (exit code 0)

Type	Name	# reqs	# fails	Avg	Min	Max	Med
		req/s	failures/s				
GET	/events?user=locust_user	112	383	180	0.62	18	0(0.00%)
Aggregated		3	180	0.62	18	0(0.00%)	183

Response time percentiles (approximated)

Type	Name	50%	66%	75%	80%	90%	95%	98%	99%	99.9%	100%	# reqs
GET	/events?user=locust_user	250	380	380	380	380	380	380	380	380	380	18
Aggregated		380	380	380	380	380	380	380	380	380	380	18

(.venv) mona@MonasLaptop:~/PES2UG23CS906/CC Lab-2\$

AFTER

FileEditViewHistoryBookmarksProfilesToolsHelp

mona@MonasLapto

mona@MonasLapto

http://localhost:8089

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LOCUST

Host: http://localhost:8000, Status: STOPPED, RPS: 0.7, Failures: 0%

STATISTICSCHARTSFAILURESEXCEPTIONSCURRENT RATIODOWNLOAD DATALOGS

STATISTICS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/events?user=locust_user	21	0	7	13	44	8.71	4	44	21138	0.7	0
Aggregated		21	0	7	13	44	8.71	4	44	21138	0.7	0

NEWRESET


```
mona@MonasLaptop X mona@MonasLaptop X + v - □ X
[2026-01-20 15:00:53,713] MonasLaptop/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 15:00:53,714] MonasLaptop/INFO/locust.main: Starting web interface at http://0.0.0.0:8089, press enter to open your default browser.
[2026-01-20 15:01:21,669] MonasLaptop/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 15:01:21,670] MonasLaptop/INFO/locust.runners: All users spawned: {"EventsUser": 1} (1 total users)
KeyboardInterrupt
2026-01-20T09:32:41Z
[2026-01-20 15:02:41,894] MonasLaptop/INFO/locust.main: Shutting down (exit code 0)
Type      Name      # reqs      # fails | Avg      Min      Max      M
ed |      req/s    failures/s
-----|-----|-----|-----|-----|-----|-----|
GET      /events?user=locust_user      21      0(0.00%) |      8
4      43      7 |      0.70      0.00
-----|-----|-----|-----|-----|-----|
3      7 |      0.70      0.00
Aggregated      21      0(0.00%) |      8      4      4

Response time percentiles (approximated)
Type      Name      50%      66%      75%      80%      90%      95%      98%
99%      99.9%      99.99%      100% # reqs
-----|-----|-----|-----|-----|-----|-----|-----|
GET      /events?user=locust_user      7      7      8      8
10      13      44      44      44      44      44      21
-----|-----|-----|-----|-----|-----|-----|-----|
44      44      44      44      44      21
Aggregated      7      7      8      8      10      13

(.venv) mona@MonasLaptop:~/PES2UG23CS906/CC Lab-2$
```

FileEditViewHistoryBookmarksProfilesToolsHelp

mona@MonasLaptop

http://localhost:8089

Task ListTechjunkie ToolsTip of My Tongue - Ch...FunSchoolOther Bookmarks

LOCUST

STATISTICSCHARTSFAILURES EXCEPTIONS CURRENT

STATISTICS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)
GET	/events?user=locust_user	21	0	7	13	44
Aggregated		21	0	7	13	44

ABOUT

mona@MonasLaptop

```
[2026-01-20 15:00:53,713] MonasLaptop/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 15:00:53,714] MonasLaptop/INFO/locust.main: Starting web interface at http://0.0.0.0:8089, press enter to open your default browser.
[2026-01-20 15:01:21,669] MonasLaptop/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 15:01:21,670] MonasLaptop/INFO/locust.runners: All users spawned: {"EventsUser": 1} (1 total users)
KeyboardInterrupt
2026-01-20T09:32:41Z
[2026-01-20 15:02:41,894] MonasLaptop/INFO/locust.main: Shutting down (exit code 0)
Type      Name      # reqs      # fails      Avg      Min      Max      Med
| req/s    failures/s
-----|-----|-----|-----|-----|-----|
GET      /events?user=locust_user      21      0(0.00%)      8
4      43      7      0.70      0.00
-----|-----|-----|-----|-----|
Aggregated      21      0(0.00%)      8      4      4
3      7      0.70      0.00

Response time percentiles (approximated)
Type      Name      50%      66%      75%      80%      90%      95%      98%
99%      99.9%      99.99%      100%      # reqs
-----|-----|-----|-----|-----|-----|-----|-----|
GET      /events?user=locust_user      7      7      8      8
10      13      44      44      44      44      44      21
-----|-----|-----|-----|-----|-----|-----|
Aggregated      7      7      8      8      10      13
44      44      44      44      21
```

(.venv) mona@MonasLaptop: ~/PES2UG23CS906/CC_Lab-2\$

RPS 0.6->0.7

ATM 183->8.7

What was the bottleneck? The bottleneck was inefficient computation inside the request handler. The routes were doing unnecessary loops and repeated calculations every time a user hit the endpoint, which increased response time under load.

What change did you make? Removed waste section

Why did the performance improve? Unnecessary computation and resource removed

MY_EVENTS

LOCUST

Hosthttp://localhost:8000

StatusSTOPPED

RPS0.5

Failures0%

NEW

RESET

STATISTICSCHARTSFAILURES EXCEPTIONS CURRENT RATIO DOWNLOAD DATA LOGS

STATISTICS

STATISTICS

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/my-events?user=locust_user	19	0	77	120	120	79.69	62	123	3144	0.5	0
Aggregated		19	0	77	120	120	79.69	62	123	3144	0.5	0

ABOUT

```
mona@MonasLaptop x mona@MonasLaptop x + v - □ X
[2026-01-20 15:04:09,705] MonasLaptop/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 15:04:09,706] MonasLaptop/INFO/locust.main: Starting web interface at http://0.0.0.0:8089, press enter to open your default browser.
[2026-01-20 15:04:40,364] MonasLaptop/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 15:04:40,365] MonasLaptop/INFO/locust.runners: All users spawned: {"MyEventsUser": 1} (1 total users)
KeyboardInterrupt
2026-01-20T09:35:23Z
[2026-01-20 15:05:23,699] MonasLaptop/INFO/locust.main: Shutting down (exit code 0)
Type      Name      # reqs      # fails | Avg      Min      Max      M
ed |      req/s    failures/s
-----|-----|-----|-----|-----|-----|-----|
GET      /my-events?user=locust_user      19      0(0.00%) |
79      62      122      77 |      0.64      0.00
-----|-----|-----|-----|-----|-----|
-----|-----|
2      77 |      0.64      0.00
Aggregated      19      0(0.00%) |      79      62      12

Response time percentiles (approximated)
Type      Name      50%      66%      75%      80%      90%      95%      98%
99%      99.9%      99.99%      100% # reqs
-----|-----|-----|-----|-----|-----|-----|
GET      /my-events?user=locust_user      77      80      86
92      110      120      120      120      120      120      19
-----|-----|-----|-----|-----|-----|
-----|-----|
Aggregated      77      80      86      92      110      120
120      120      120      120      120      19

(.venv) mona@MonasLaptop:~/PES2UG23CS906/CC Lab-2$
```

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mona@MonasLapto

mona@MonasLapto

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LOCUST

STATISTICSCHARTSFAILURESEXCEPTIONSCURRENT

⏏

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)
GET	/my-events?user=locust_user	19	0	77	120	120
Aggregated		19	0	77	120	120

ABOUT

[2026-01-20 15:04:09,705] MonasLapto/INFO/locust.main: Starting Locust 2.43.1

[2026-01-20 15:04:09,706] MonasLapto/INFO/locust.main: Starting web interface at http://0.0.0.0:8089, press enter to open your default browser.

[2026-01-20 15:04:40,364] MonasLapto/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second

[2026-01-20 15:04:40,365] MonasLapto/INFO/locust.runners: All users spawned: {"MyEventsUser": 1} (1 total users)

KeyboardInterrupt

[2026-01-20 15:05:23,699] MonasLapto/INFO/locust.main: Shutting down (exit code 0)

Type | Name | # reqs | # fails | Avg | Min | Max | Med | req/s | failures/s

-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|

GET | /my-events?user=locust_user | 19 | 0 | 77 | 120 | 120 | 79 | 62 | 122 | 77 | 0.64 | 0.00

-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|

Aggregated | 19 | 0 | 77 | 120 | 120 | 2 | 77 | 0.64

-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|

Response time percentiles

Type | Name | 50% | 90% | 95% | 99% | 99.9%

-----|-----|-----|-----|-----|-----|-----|

GET | /my-events?user=locust_user | 120 | 120 | 120 | 120 | 120

-----|-----|-----|-----|-----|-----|-----|

Aggregated | 120 | 120 | 120 | 120 | 120

-----|-----|-----|-----|-----|-----|-----|

(.venv) mona@MonasLapto:~

mona@MonasLapto:~

[2026-01-20 15:04:09,705] MonasLapto/INFO/locust.main: Starting Locust 2.43.1

[2026-01-20 15:04:09,706] MonasLapto/INFO/locust.main: Starting web interface at http://0.0.0.0:8089, press enter to open your default browser.

[2026-01-20 15:04:40,364] MonasLapto/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second

[2026-01-20 15:04:40,365] MonasLapto/INFO/locust.runners: All users spawned: {"MyEventsUser": 1} (1 total users)

KeyboardInterrupt

[2026-01-20 15:05:23,699] MonasLapto/INFO/locust.main: Shutting down (exit code 0)

Type | Name | # reqs | # fails | Avg | Min | Max | Med | req/s | failures/s

-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|

GET | /my-events?user=locust_user | 19 | 0 | 77 | 120 | 120 | 79 | 62 | 122 | 77 | 0.64 | 0.00

-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|

Aggregated | 19 | 0 | 77 | 120 | 120 | 2 | 77 | 0.64

-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|

Response time percentiles

Type | Name | 50% | 90% | 95% | 99% | 99.9%

-----|-----|-----|-----|-----|-----|-----|

GET | /my-events?user=locust_user | 120 | 120 | 120 | 120 | 120

-----|-----|-----|-----|-----|-----|-----|

Aggregated | 120 | 120 | 120 | 120 | 120

-----|-----|-----|-----|-----|-----|-----|

Snipping Tool

Screenshot copied to clipboard

Automatically saved to screenshots folder.

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AFTER OPTIMIZATION

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Monolithic Applications

Locust

PES2UG23CS906 - C

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←→↻http://localhost:8089

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LOCUST

Host: http://localhost:8000

Status: STOPPED

RPS: 0.5

Failures: 0%

NEW

RESET

⚙

STATISTICSCHARTSFAILURESEXCEPTIONSCURRENT RATIODOWNLOAD DATALOGS

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Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/my-events?user=locust_user	20	0	7	58	58	8.52	3	58	3144	0.5	0
Aggregated		20	0	7	58	58	8.52	3	58	3144	0.5	0

```
mona@MonasLaptop: ~$ python3 -m locust
[2026-01-20 15:06:27,535] MonasLaptop/INFO/locust.main: Starting Locust 2.43.1
[2026-01-20 15:06:27,536] MonasLaptop/INFO/locust.main: Starting web interface at http://0.0.0.0:8089, press enter to open your default browser.
[2026-01-20 15:07:06,867] MonasLaptop/INFO/locust.runners: Ramping to 1 users at a rate of 1.00 per second
[2026-01-20 15:07:06,868] MonasLaptop/INFO/locust.runners: All users spawned: {"MyEventsUser": 1} (1 total users)
KeyboardInterrupt
2026-01-20T09:38:09Z
[2026-01-20 15:08:10,000] MonasLaptop/INFO/locust.main: Shutting down (exit code 0)
Type      Name      # reqs      # fails | Avg      Min      Max      M
ed |      req/s    failures/s
-----|-----|-----|-----|-----|-----|-----|
GET      /my-events?user=locust_user      20      0(0.00%) |
8      3      58      7 |      0.69      0.00
-----|-----|-----|-----|-----|-----|
Aggregated      20
8      7 |      0.69
Response time percentiles (
Type      Name      50%      6
99%      99.9%      99.99%      100
-----|-----|-----|-----|
GET      /my-events?user=lo
7      8      58      58
-----|-----|-----|-----|
Aggregated
58      58      58      58
(.venv) mona@MonasLaptop: ~$
```

mona@MonasLaptop

mona@MonasLaptop

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[2026-01-20 15:05:23,699] MonasLaptop/INFO/locust.main: Shutting down (exit code 0)

Type Name # reqs # fails | Avg Min Max M
ed | req/s failures/s

-----|-----|-----|-----|-----|-----|-----|

GET /my-events?user=locust_user 19 0(0.00%) |

79 62 122 77 | 0.64 0.00

-----|-----|-----|-----|-----|-----|-----|

Aggregated 19 0(0.00%) |

2 77 | 0.64 0.00

-----|-----|-----|-----|-----|-----|-----|

Response time percentiles (approximated)

Type Name 50% 66% 75% 90% 95% 99%

-----|-----|-----|-----|-----|-----|-----|

GET /my-events?user=lo

7 8 58 58

-----|-----|-----|-----|-----|-----|-----|

Aggregated

58 58 58 58

-----|-----|-----|-----|-----|-----|-----|

Snipping Tool

Screenshot copied to clipboard

Automatically saved to screenshots folder.

Markup and share

The screenshot displays the Locust web interface in a browser window and the corresponding terminal output. The browser window shows the Locust logo and navigation tabs: STATISTICS, CHARTS, FAILURES, EXCEPTIONS, and CURRENT. The terminal window shows the Locust 2.43.1 logs, including the start of the web interface at http://0.0.0.0:8089, the spawning of 1 user, and the shutdown of the process. The terminal also displays the Locust statistics for the GET request to /my-events?user=locust_user, which matches the data shown in the browser interface.

Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)
GET	/my-events?user=locust_user	20	0	7	58	58
Aggregated		20	0	7	58	58

RPS 0.64->0.69

ATM 73->8

Same logic used here

What was the bottleneck? The bottleneck was inefficient computation inside the request handler. The routes were doing unnecessary loops and repeated calculations every time a user hit the endpoint, which increased response time under load.

What change did you make? Removed dummy iteration section

Why did the performance improve? Unnecessary computation and resource removed